



机器学习导论

第2章 线性模型 2.2 Logistic回归

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Logistic回归部分的内容主要来自于吴恩达Coursera网课

什么是分类(Classification)问题?

垃圾邮件检测任务: Spam / Not Spam?

信用卡交易欺诈检测: Fraudulent (Yes / No)?

肿瘤判定: Malignant (恶性的) / Benign (良性的)?

$y \in \{0, 1\}$ 0: 负类 (Negative Class) , 有的方法也标注为-1

1: 正类 (Positive Class)

回归和分类任务的区别1

- 输入相同，输出类型不同

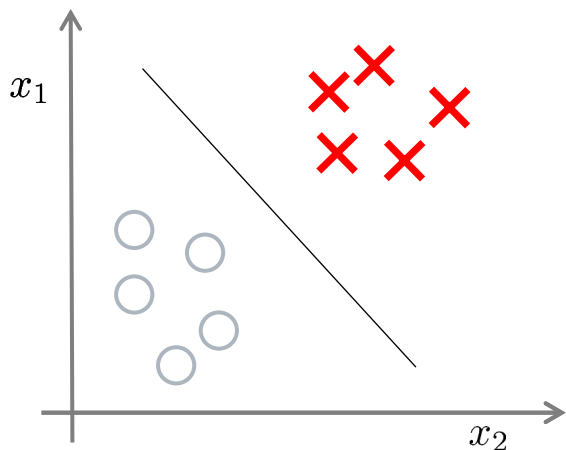
分类: $h_{\theta}(x) = 0 \text{ or } 1$ (类别)

回归: $h_{\theta}(x) \in \mathbb{R}$ (连续)

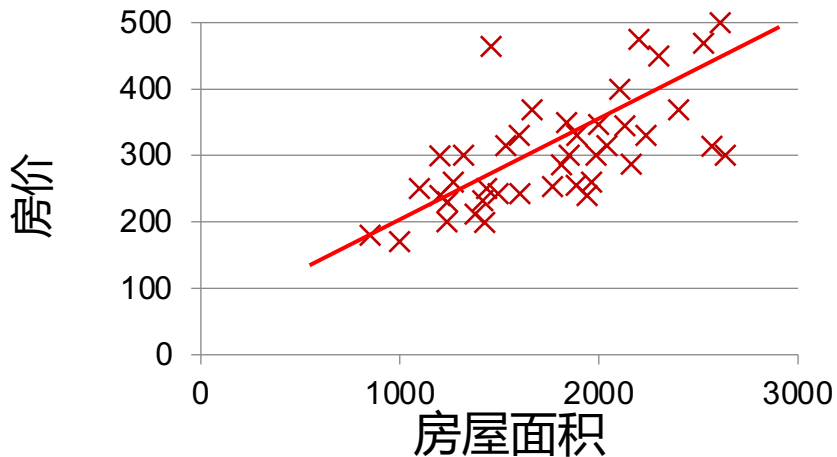
- 分类适用范围更广一些
 - Logistic Regression, LDA, SVM, ANN, 决策树, K最近邻...

线性回归和分类的区别 2

分类



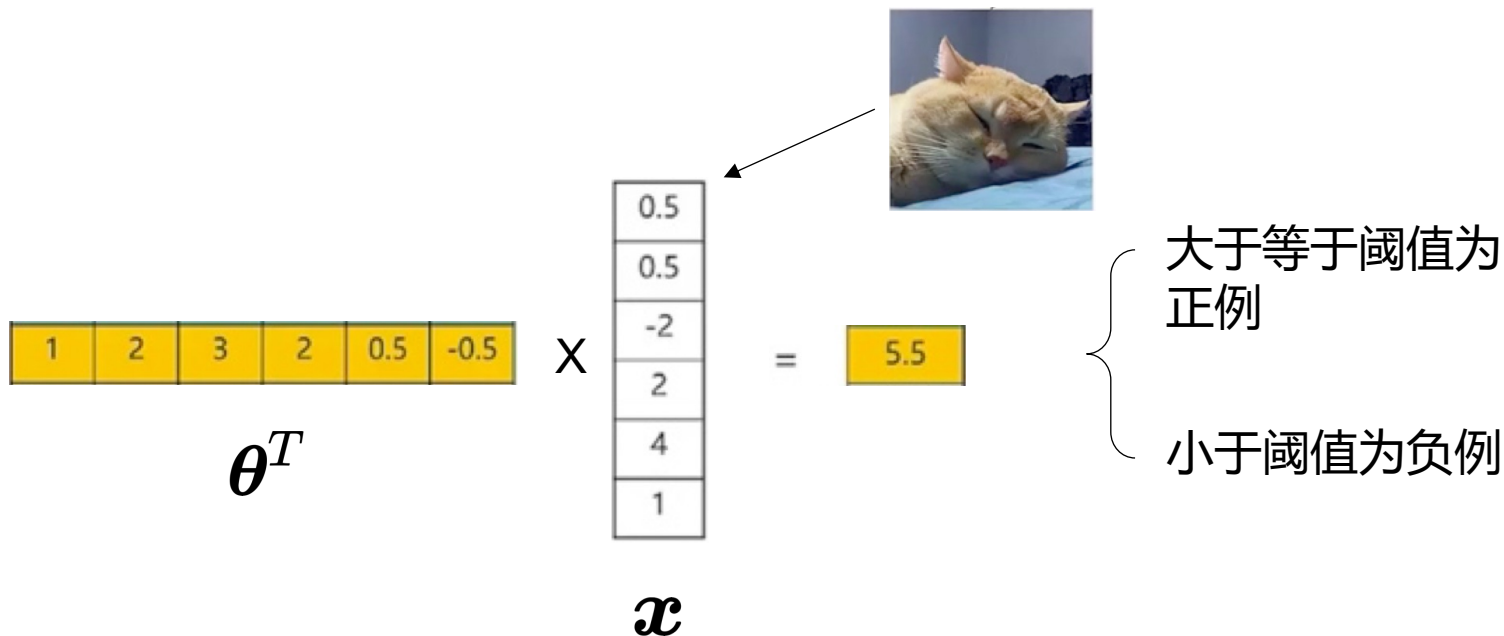
回归



判决函数的形式虽都为 $h_{\theta}(x) = \theta^T x$ ，但对结果的使用上不同。

什么是线性分类器?

- 分类器的判决函数主体形式为: $h_{\theta}(x) = \theta^T x$



01. Logistic回归

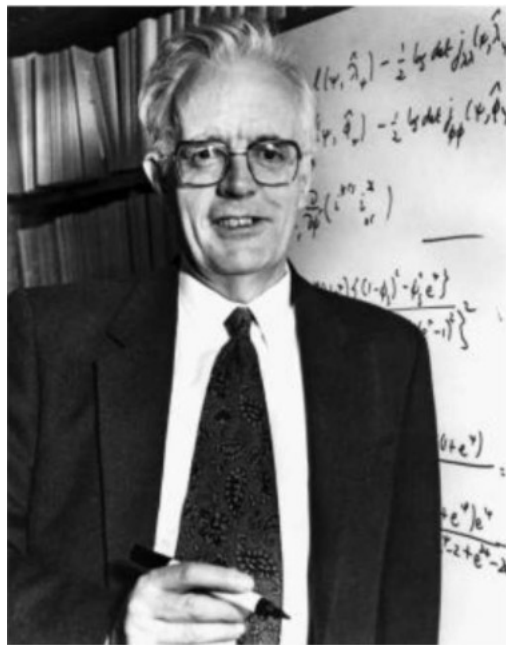
- 模型表示与判别函数
- 代价函数
- 基于梯度下降法的模型优化

02. 线性判别分析(Linear Discriminative Analysis)

- 模型表示与判别函数
- 代价函数
- 基于梯度下降法的模型优化

Logistic回归简介

- Logistic Regression由统计学家David Cox于1958年提出。
- D. Cox最著名的成果是1972年的Cox Regression，广泛用于生存预测和机械故障等



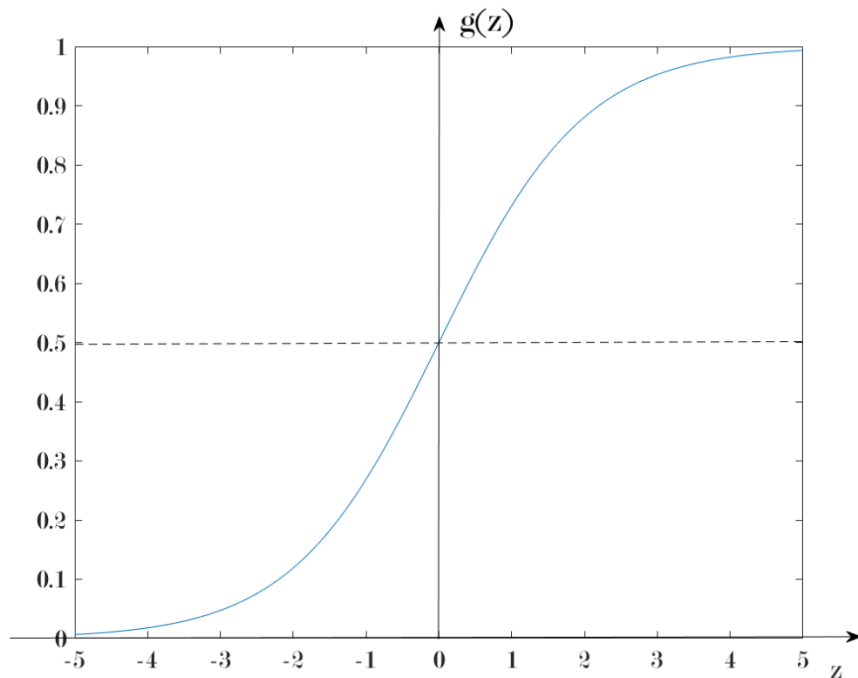
David Cox(1924-2022)

Logistic回归模型

- 主要使用sigmoid函数, 将线性输出 $\theta^T x$ 大部分变换到0或1附近
- Logistic回归模型主体:

$$h_{\theta}(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}}$$

其中, $g(z) = \frac{1}{1 + e^{-z}}$



1.1 Logistic回归模型的判决函数

$$h_{\theta}(x) = g(\theta^T x)$$

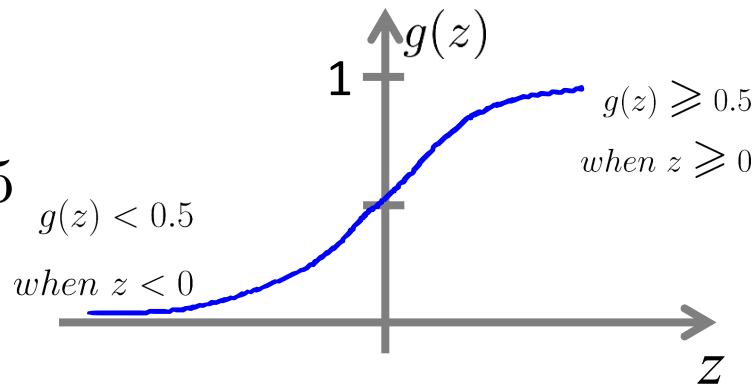
$$g(z) = \frac{1}{1+e^{-z}}$$

模型判决“ $y = 1$ ”时 $h_{\theta}(x) \geq 0.5$

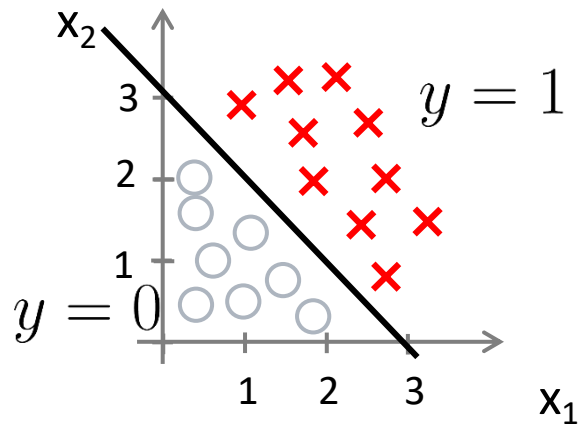
对应地, $\theta^T x \geq 0$

反之, 当“ $y = 0$ ”时 $h_{\theta}(x) < 0.5$

$$\theta^T x < 0$$



1.1 判决边界或判决函数(Decision Boundary, Prediction Model)



判决边界或预测模型:

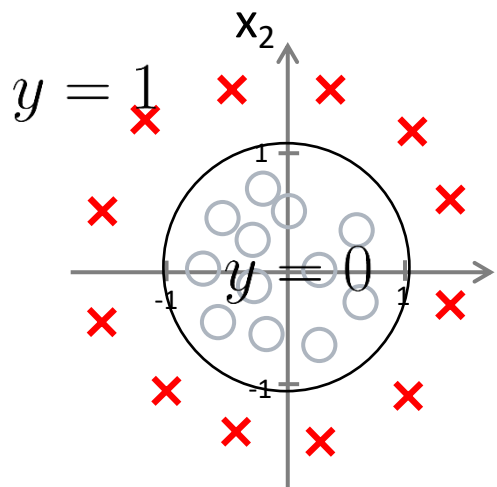
$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

在示例中: $\theta_0 = -3, \theta_1 = 1, \text{ and } \theta_2 = 1$.

预测 “ $y = 1$ ” 当 $-3 + x_1 + x_2 \geq 0$

预测 “ $y = 0$ ” 当 $-3 + x_1 + x_2 < 0$

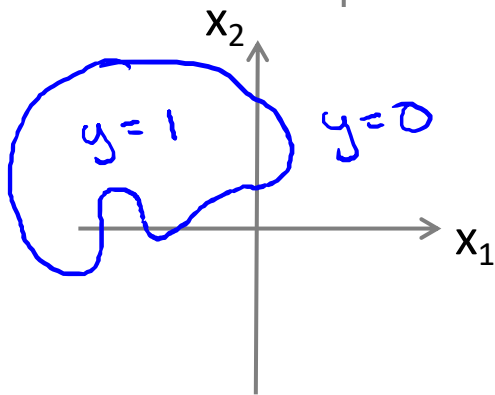
1.1 非线性判决函数



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2) \quad \theta = \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

Predict “ $y = 1$ ” if $-1 + x_1^2 + x_2^2 \geq 0$

$$x_1^2 + x_2^2 \geq 1$$



$$h_{\theta}(x) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_1^2 x_2 + \theta_5 x_1^2 x_2^2 + \theta_6 x_1^3 x_2 + \dots)$$

1.2 Logistic回归的代价函数

Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})\}$

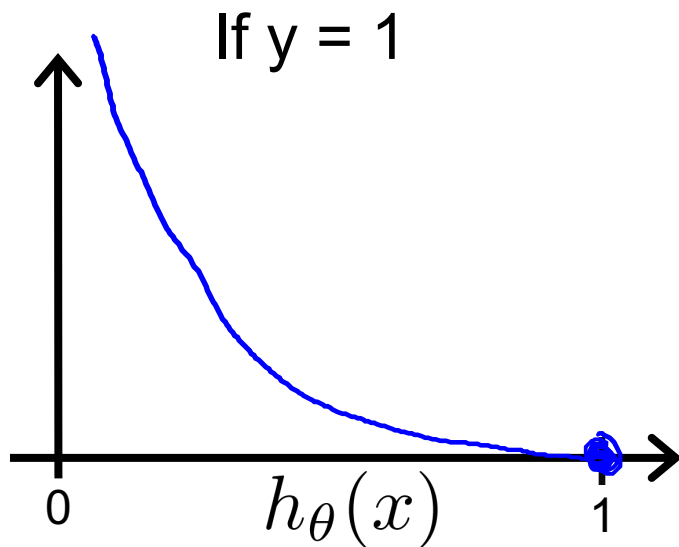
m examples $x \in \begin{bmatrix} x_0 \\ x_1 \\ \dots \\ x_n \end{bmatrix}$, $x_0 = 1, y \in \{0, 1\}$
where

Model of
classification:
$$h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

How to choose parameters θ ?

1.2 Logistic regression cost function

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



Cost = 0 if $y = 1, h_{\theta}(x) = 1$

But as $h_{\theta}(x) \rightarrow 0$

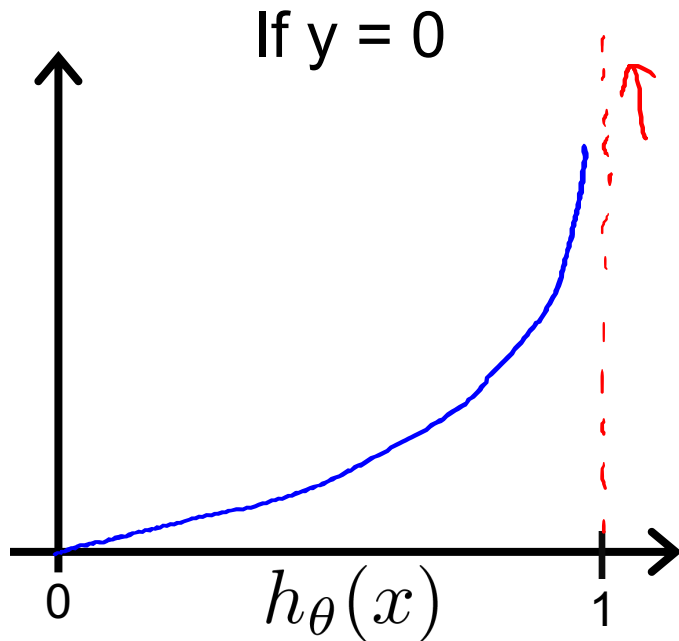
$\text{Cost} \rightarrow \infty$

Captures intuition that if $h_{\theta}(x) = 0$,
(predict $P(y = 1|x; \theta) = 0$), but $y = 1$,
we'll penalize learning algorithm by a very
large cost.

Logistic regression cost function



$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



1.2 Logistic regression cost function

$$\begin{aligned} J(\theta) &= \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_{\theta}(x^{(i)}), y^{(i)}) \\ &= -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right] \end{aligned}$$

$$\text{Cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$

注意： $y^{(i)}$ 是样本的标注，只能取值0或1

- 目标：通过 $\min_{\theta} J(\theta)$ 来获得最拟合训练样本的模型参数 θ

1.2 损失函数 - 交叉熵

交叉熵主要用于度量两组数据分布间的差异性

$$CrossEnt(P, Q) = \sum_{k=1}^{|\mathcal{Y}|} p_k \log_2 \frac{1}{q_k} = - \sum_{k=1}^{|\mathcal{Y}|} p_k \log_2 q_k$$

$$CrossEnt(P, Q) = \int_x P(x) \log_2 \frac{1}{Q(x)} dx = \mathbb{E}_p(-\log_2 Q(x))$$

1.2 损失函数 - 交叉熵-信息熵

信息论：香农(Shannon)

信息熵 (Information Entropy) 是度量样本集合纯度的一种指标。假定当前样本集合 D 中第 k 类样本所占的比例为 p_k ($k = 1, 2, \dots, |Y|$)，则 D 的信息熵定义为：

$$Ent(D) = \sum_{k=1}^{|Y|} p_k \log_2 \frac{1}{p_k} = - \sum_{k=1}^{|Y|} p_k \log_2 p_k$$

$Ent(D)$ 的值越小，则 D 的纯度越高。

见周志华《机器学习》4.2节(P75)

1.3 优化(最小化代价函数):Gradient Descent

$$J(\theta) = -\frac{1}{m} \left[\sum_{i=1}^m y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)})) \right]$$

Want $\min_{\theta} J(\theta)$:

Repeat {

$$\theta_j := \theta_j - \alpha \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)} \quad \text{其中, } h_{\theta}(x) = \frac{1}{1 + e^{-\theta^T x}}$$

}

(simultaneously update all θ_j)

注意Logistic回归和线性回归的梯度的相似之处和区别

线性回归的 $h_{\theta}(x) = \theta^T x$

附：代价函数相对于模型参数的导数求导过程（共2页）

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{\partial}{\partial \theta_j} \left\{ \frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log_2 h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log_2 (1 - h_{\theta}(x^{(i)})) \right] \right\}$$

$$\begin{aligned} \frac{\partial h_{\theta}(x)}{\partial \theta} &= \frac{\partial}{\partial \theta} \frac{1}{1 + e^{-\theta x}} \\ &= - \frac{1}{(1 + e^{-\theta x})^2} \frac{\partial}{\partial \theta} (1 + e^{-\theta x}) \\ &= - \frac{1}{(1 + e^{-\theta x})^2} \cdot e^{-\theta x} \cdot \frac{\partial}{\partial \theta} (-\theta x) \\ &= - \frac{1}{(1 + e^{-\theta x})^2} \cdot e^{-\theta x} \cdot (-x) \\ &= \frac{1}{1 + e^{-\theta x}} \cdot \frac{e^{-\theta x}}{1 + e^{-\theta x}} \cdot x \\ &= h_{\theta}(x) \cdot (1 - h_{\theta}(x)) \cdot x \end{aligned}$$

$$\begin{aligned}
 \frac{\partial J(\theta)}{\partial \theta_j} &= \frac{\partial}{\partial \theta_j} \left\{ \frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \log_2 h_\theta(x^{(i)}) + (1 - y^{(i)}) \log_2(1 - h_\theta(x^{(i)})) \right] \right\} \\
 &= \frac{1}{m} \sum_{i=1}^m \left[y^{(i)} \frac{\partial}{\partial \theta_j} \log_2 h_\theta(x^{(i)}) + (1 - y^{(i)}) \frac{\partial}{\partial \theta_j} \log_2(1 - h_\theta(x^{(i)})) \right] \\
 &= \frac{1}{m} \sum_{i=1}^m \left[\frac{y^{(i)}}{h_\theta(x^{(i)}) \ln 2} \frac{\partial}{\partial \theta_j} h_\theta(x^{(i)}) + \frac{1 - y^{(i)}}{(1 - h_\theta(x^{(i)})) \ln 2} \frac{\partial}{\partial \theta_j} (1 - h_\theta(x^{(i)})) \right] \\
 &= \frac{1}{m \ln 2} \sum_{i=1}^m \left[\frac{y^{(i)}}{h_\theta(x^{(i)})} \frac{\partial}{\partial \theta_j} h_\theta(x^{(i)}) + \frac{1 - y^{(i)}}{1 - h_\theta(x^{(i)})} \frac{\partial}{\partial \theta_j} (-h_\theta(x^{(i)})) \right] \\
 &= \frac{1}{m \ln 2} \sum_{i=1}^m \left[\left(\frac{y^{(i)}}{h_\theta(x^{(i)})} - \frac{1 - y^{(i)}}{1 - h_\theta(x^{(i)})} \right) \frac{\partial}{\partial \theta_j} h_\theta(x^{(i)}) \right] \\
 &= \frac{1}{m \ln 2} \sum_{i=1}^m \left(\frac{y^{(i)} - h_\theta(x^{(i)})}{h_\theta(x^{(i)})(1 - h_\theta(x^{(i)}))} \frac{\partial}{\partial \theta_j} h_\theta(x^{(i)}) \right) \\
 &= \frac{1}{m \ln 2} \sum_{i=1}^m (y^{(i)} - h_\theta(x^{(i)})) x^{(i)}
 \end{aligned}$$

- 二类分类
- 模型表示与判别函数
- 基于交叉熵的损失函数
- 基于梯度下降的优化算法

D. Cox使用 $y = \frac{1}{1 + e^{-\theta^T \mathbf{x}}}$ 拟合二类分类数据, 经过变换
可得 $\theta^T \mathbf{x} = \ln \frac{y}{1 - y}$ 亦被称为对数几率(log odds或logit), 可以表明类别归属
将y视为类后验概率估计, 可得 $P(y = 1 | \mathbf{x}) = \frac{e^{\theta^T \mathbf{x}}}{1 + e^{\theta^T \mathbf{x}}}$
 $P(y = 0 | \mathbf{x}) = \frac{1}{1 + e^{\theta^T \mathbf{x}}}$

使用最大对数似然和训练数据集估计模型参数 θ

$$\theta = \arg \max_{\theta} \sum_{i=1}^m \ln P(y_i | \mathbf{x}_i; \theta)$$